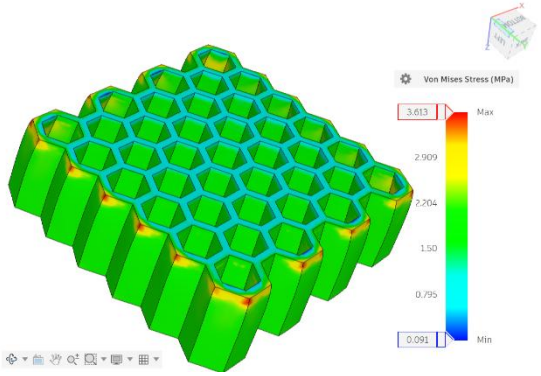


Midsole Lattice Optimisation

Core objective: Parametric comparison of three TPU lattice geometries under an 80 kg runner heel-strike load (2,000 N) to maximise Specific Energy Absorption (SEA) while reducing weight.

Table of key information

	Solid Baseline block	Re-entrant Auxetic (bowtie)	Hexagonal Honeycomb	Diamond Grid
TPU Mass (g)	60	23.97	23.31	23.3
Mass reduction (%)	0	60.1	61.2	61.2
Comparison of how well the designs perform against each other:				
Peak stress (MPa)	-	3.821	3.613	4.447
Deformation in direction of load (mm)	-	1.412	1.473	1.458
Stiffness (N/mm)	-	1416	1358	1372
SEA (J/kg)	-	58.9	63.2	62.6



- **Performance Winner**: Re-entrant Auxetic delivers highest stiffness (1,416.43 N/mm), whereas Honeycomb yields the highest Specific Energy Absorption (63.19 J/kg).
- **FEA Singularity & Stress Concentration**: Fixed bottom boundary corners concentrate maximum stress. Un-filleted internal re-entrant joints induce a corner singularity in linear static FEA; add $R = 0.5$ mm fillets in future iterations to mitigate fatigue.
- **Mesh Independence**: A 1.5 mm element baseline was validated against 3.0 mm and 1.0 mm meshes to balance solve speed and bending resolution.

Design for Additive Manufacturing Validation

- **Printability**: 90°draft angles eliminate internal supports entirely. Uniform 1.5 mmwalls map directly to 3–4 perimeter passes on a 0.4 mmnozzle.
- **Slicing Validation**: Kiri:Moto slicer audits confirm print times of 4h 45m to 5h 00m utilizing ≈ 26g of TPU filament per unit.

Scan or click here for raw CAD files, FEA simulation datasets, and engineering drawing PDFs